

MODEL OPTIMIZATION REPORT

Advanced Regression Model Comparison & Optimization Platform

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1. Dataset Information

Property	Value
Dataset	House Prices
Total Samples	1458
Total Features	265
Target Variable	SalePrice
Train/Test Split	80% / 20%

2. Optimization Technique

RandomizedSearchCV was used to optimize the regression models. Multiple combinations of hyperparameters were evaluated using 5-fold cross-validation. The best parameter set was selected based on the highest R² Score.

3. Optimized Models

Algorithm	Optimization Method
Decision Tree	RandomizedSearchCV
Random Forest	RandomizedSearchCV
Gradient Boosting	RandomizedSearchCV
XGBoost	RandomizedSearchCV
LightGBM	RandomizedSearchCV
CatBoost	RandomizedSearchCV

4. Final Selected Model

Algorithm: CatBoost Regressor
R² Score: 0.9307
MAE: 14033.41
RMSE: 19560.91
Cross Validation: 0.9280

5. Conclusion

Hyperparameter optimization using RandomizedSearchCV significantly improved the predictive performance of the regression models. Among all optimized models, CatBoost Regressor achieved the best balance between prediction accuracy and generalization capability. Therefore, CatBoost was selected as the final deployment model.